Mobile Robotic System for Precision Surface Marking

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***Abstract*-- This project develops an autonomous robot capable of drawing figures on flat surfaces, aimed at applications in construction and woodworking. The system utilizes a TurtleBot and a one-degree-of-freedom mechanism with a marking tool, controlled via pseudo-G code and a PD/PID control system to ensure precision. Expected results include improved accuracy and reduced errors compared to manual drawing, enabling direct marking on construction sites and better material optimization in woodworking. The significance of this project lies in its potential to reduce time and costs, minimize waste, and enhance efficiency in marking tasks, thereby contributing to industrial automation. This document presents the methodology for mechanical design, electronics, programming, and control, as well as the final schematics of the marking mechanism that attaches to the TurtleBot and allows it to write.**

I. INTRODUCTION

The current challenges in site marking and material cutting significantly impact accuracy, efficiency, and costs in construction and manufacturing projects. To better understand these issues and evaluate the feasibility of the proposed robot, we conducted an interview with Mr. Segundo Ávila, owner of Astoria Custom WoodWork in Brooklyn, NY, a relative of one of the team members. Additionally, we reviewed literature on robots performing similar tasks, such as the HP SitePrint [1], the HK1500 from SuperDroid Robots [2], and a micro-ROS-based robot using a marker.

Manual marking faces challenges in precision and accuracy, particularly in construction, where errors in marking long lines or complex figures can cause misalignments in structural phases, affecting element arrangement. Similarly, errors in material cutting (e.g., wood) lead to incorrect cuts, resulting in material waste and additional costs, especially when working with expensive materials or designs requiring maximum material utilization.

Achieving manual precision demands time and skilled workers, increasing costs and limiting productivity in large projects. Worker fatigue during prolonged marking sessions further increases the likelihood of errors. In material cutting, every line must be marked before cutting, making the process labor-intensive. Automation can significantly reduce operation time, allowing workers to focus on higher-value tasks.

Errors in marking lead to elevated costs in construction and manufacturing. In construction, inaccurate marking causes rework, material waste, and delays. In manufacturing, errors result in material adjustment or discard, increasing costs and production time.

Beneficiary: Mr. Segundo Ávila is the owner of Astoria Custom WoodWork

Benefits: The robot can reduce errors and rework by eliminating precision variations and common manual marking mistakes. This translates to less rework and material waste. Automation also allows marking tasks to be completed faster than manual methods, especially for complex designs. Workers can focus on tasks requiring human intervention, improving project efficiency.

II. PRIMARY RESEARCH

A. Interview with Mr. Segundo Ávila

Mr. Segundo Ávila is the owner of Astoria Custom WoodWork, a small carpentry, construction, and apartment remodeling company based in Brooklyn, NY. The company specializes in high-quality, custom carpentry and construction projects, primarily catering to a demanding clientele in Manhattan. Due to the nature of their work, they place significant emphasis on accuracy, quality, and timely delivery to maintain their reputation in this competitive market.

Astoria Custom WoodWork operates with a small core team of skilled professionals, including Mr. Segundo Ávila and Mr. Héctor Ávila, both of whom have extensive experience in the field. This expertise enables them to consistently deliver projects of exceptional quality. However, their operational model is highly influenced by the seasonal nature of the construction and carpentry industry.

During peak seasons, such as summer, the company secures multiple contracts simultaneously, often requiring an expanded workforce. Conversely, during low-demand periods, such as winter, the company experiences a significant reduction in contracts, sometimes leading to no active projects at all. To adapt to this variability, the company employs a flexible workforce model, relying heavily on a trusted group of five skilled workers. However, availability issues among these trusted professionals often force the company to hire external workers.

The reliance on external hires introduces several risks and inefficiencies. According to Mr. Ávila, finding workers with the required level of skill and expertise is a recurring challenge. Critical tasks such as plan interpretation, accurate marking, and

precise measurement require a level of attention to detail and technical understanding that many temporary workers lack.

B. Common Mistakes

- Misinterpretation of plans or distractions.
- Incorrect equipment usage or leveling.
- Overstated experience without essential practical skills.

This causes some economic problems for the business models, meaning in a loss given the fact that you must rework the pieces correctly having an unexpected price tag added for wasted materials or incorrectly marked pieces. Also, delays caused by rework and misalignments increase project durations, the daily labor is around 150-400$ per day alongside potential penalties.
Poorly marked plans affect the structural and aesthetic quality of the final product. Misaligned walls or improperly placed doors and windows reduce functionality and can tarnish the company's reputation.

To mitigate these challenges, Astoria Custom WoodWork employs several strategies.

Mr. Segundo and Mr. Héctor personally oversee critical stages of each project, particularly the initial marking phase. Their direct involvement helps identify and correct errors early, reducing the likelihood of costly rework.
Before initiating projects, workers receive detailed instructions on task-specific procedures. This ensures that even temporary workers have a clear understanding of their responsibilities, reducing errors caused by lack of knowledge.
While these measures have proven effective to some extent, they come with limitations, the first being time consuming. Training new employees can take around one week and a month representing significant expense, especially if the employee does not remain in the company. He also mentioned that constant supervision by senior team members limits their ability to focus on other critical activities, such as acquiring new contracts or managing complex projects. This lack of delegation hinders business growth and scalability.

III. SECONDARY RESEARCH

The development of autonomous robots for marking and tracing has made significant advances in recent years. One of the most notable examples is the HP SitePrint, a robot specialized in marking architectural plans on construction sites [1]. This system uses autonomous navigation technology and can perform complex markings with high precision, significantly reducing the marking time compared to manual methods.

The *HK1500* from *SuperDroid Robots* represents another approach to the problem, focusing on larger-scale industrial applications. This robot employs a control system based on absolute coordinates and can operate on uneven surfaces thanks to its adaptive suspension system, although its precision is limited on highly rugged terrain [2].

In the academic realm, projects involving mobile robots have sought to solve various challenges in marking tasks, demonstrating the feasibility of using robotic systems for these applications.

Existing systems share common features, such as:

- Use of *LIDAR* sensors for navigation and obstacle detection.
- Implementation of *PID control algorithms* to achieve movement precision.
- *Compensation systems* for terrain irregularities.
- *CAD-based user interfaces* for trajectory programming.

The present project aims to develop a robust solution capable of operating in different marking tasks with a favorable cost-benefit ratio, making it suitable for small and medium-sized enterprises.

Nevertheless, it is important to mention that control systems are always related to this project in so many ways. We are going to mention a couple of controllers could be needed in our solutions.

Proportional-Integral (PI) and Proportional-Derivative (PD) controllers are widely used in control systems to enhance performance by reducing error and ensuring stability. PI controllers are particularly effective for systems that require long-term accuracy, as they address steady-state errors by integrating the error over time. However, they tend to respond slower to dynamic changes, making them less ideal for applications where rapid adjustments are critical. In contrast, PD controllers excel in systems requiring quick responsiveness, as they combine proportional error correction with predictive adjustments based on the rate of change. This allows PD controllers to stabilize systems efficiently while minimizing overshoot and oscillations.[5]

For this project, the choice between PI and PD control depends on the dynamic nature of maintaining consistent contact with the surface and importance of precise force adjustments. Emphasis is placed on responsiveness and stability, leading to a focus on PD control as a more suitable solution.

COMPARATIVE ANALYSIS OF EXISTING SOLUTIONS
A comparative analysis was conducted on the different solutions available in the market, focusing on aspects such as operational speed and equipment acquisition costs.

1. Operational Speed:
   - HP SitePrint: 900 m/hour
   - HK1500: 4.82 km/hour
2. Approximate Costs:
   - HP SitePrint: >$50,000 USD
   - HK1500: >$30,000 USD

MARKET OPPORTUNITY

Based on this analysis, there is a clear market opportunity for low-cost systems. The proposed prototype will aim to achieve:

- A maximum operational speed of 300 m/hour.
- A total cost of less than $2,000 USD.

IV. Problem Definition

In the field of construction, precision in marking architectural plans on-site is critical to ensure that the final structure complies with technical and design specifications. However, this task is still predominantly performed manually by technicians and builders, which introduces several challenges that negatively impact on the quality, efficiency, and cost of projects.

Manual marking is a labor-intensive process that depends heavily on the skill and experience of workers. Errors often occur due to misinterpretation of plans, fatigue, or improper equipment use. For instance, as highlighted in our interview with Mr. Segundo Ávila, owner of Astoria Custom WoodWork in Brooklyn, NY, workers sometimes misalign markings, leading to structural misplacements such as walls, doors, or windows being out of alignment. These errors are costly, with rework expenses ranging between $1,500 and $5,000 USD, and the added daily labor costs ranging from $150 to $400 USD per worker. Furthermore, clients in high-demand markets like Manhattan often penalize delays, adding further financial strain.

The seasonal nature of construction work exacerbates these challenges. Companies like Astoria Custom WoodWork frequently rely on a core team of skilled workers but must hire temporary staff during peak periods. These temporary workers often lack the expertise required for precise marking, further increasing the risk of errors. Beyond financial costs, these mistakes can compromise the aesthetic quality and functional alignment of structures, affecting client satisfaction and a company's reputation.

Additionally, manual marking is time-consuming, particularly for complex or large-scale architectural designs. Each marking must be measured and drawn individually, a process that becomes exponentially slower for intricate or repetitive patterns. The inefficiency of this method limits productivity and delays project completion, making automation a necessary advancement to address these issues.

V. Proposed Solution

The proposed solution focuses on the development of an automated system using a controlled differential robot to perform real-scale architectural plan marking. This robot will be equipped with a writing control system capable of accurately and efficiently reproducing the plans at construction sites, eliminating common errors associated with manual processes. It is anticipated that the robot will significantly reduce the time required for this stage and improve the accuracy of the marking.

The main objectives of this project are:

1. To minimize marking errors on-site, ensuring that the marked elements align with the original design.
2. To reduce the marking time compared to manual methods, which can contribute to better time management and help avoid delays in the construction process.
3. To implement autonomous controls so the robot can independently perform the marking, following the provided plans without the need for constant operator intervention. This development will advance the automation of the construction process, enhancing precision and optimizing human resources on-site.

Mechanics

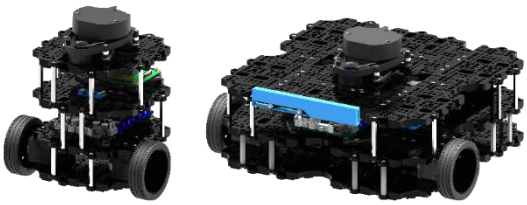

Figure 1 Turtlebot Burger and Turtlebot Waffle

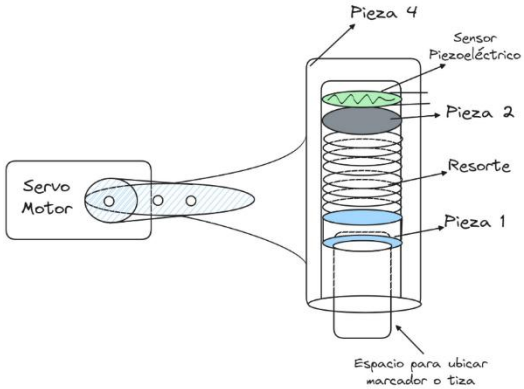

Figure 2 Proposal A

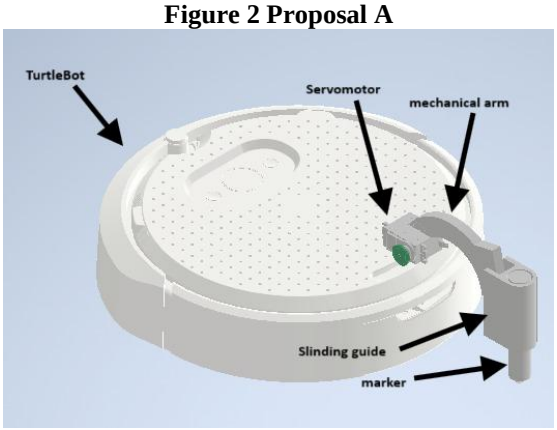

Figure 3 Proposal A attached

The mechanism shown in Figure 2 uses a servo motor to generate controlled rotational movement, which applies pressure and displacement to the marker. Piece 1 secures the marker and withstands internal forces, while the spring regulates both the applied force and linear motion. Piece 2 transmits the force between the spring and the Piezoelectric sensor, which measures applied pressure, enabling calibrations and preventing excessive pressure. Piece 4 protects the internal

components, supports the sensor, and guides the linear motion.

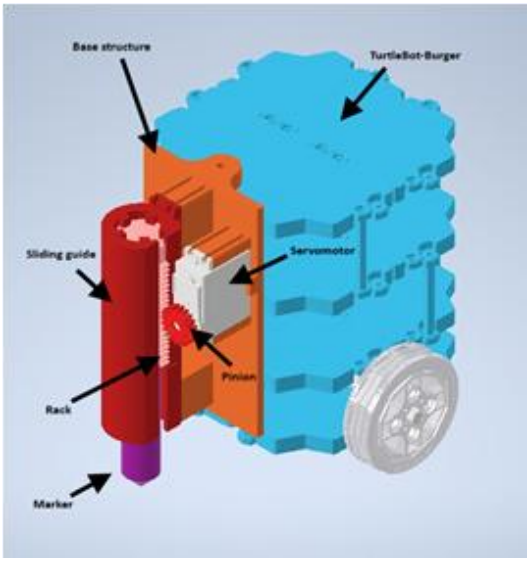

Figure 4 Mechanical Solution Proposal B

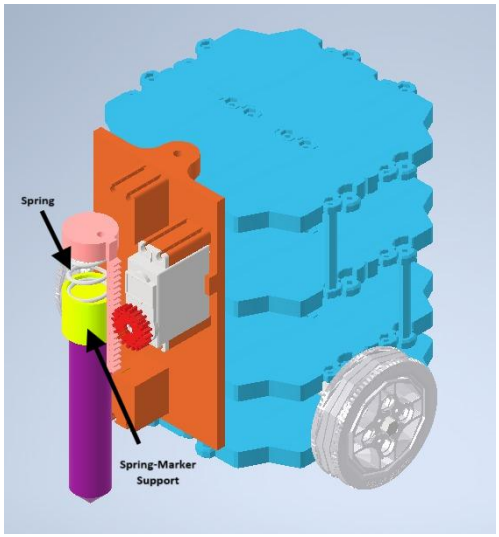

Figure 5 Mechanical Solution Proposal B (Hiding the visibility of the slider)

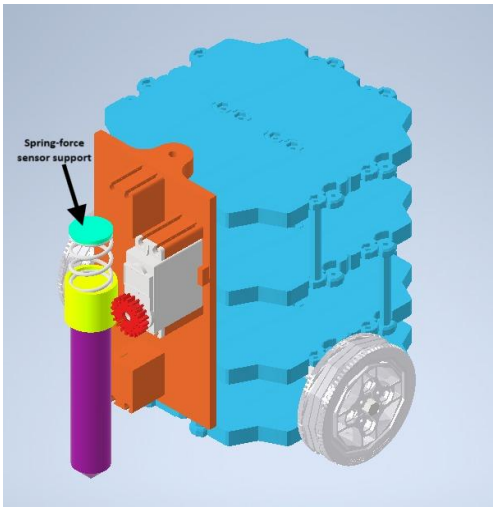

Figure 6 Mechanical Solution Proposal B (Hiding the visibility of the rack)

The mechanism consists of a fixed base that connects the TurtleBot to a slider and a servo motor. A pinion, coupled to the servo motor's shaft, transmits power to a vertically moving rack. At its end, a plastic piece holds a spring and a piezoelectric sensor that measures the applied pressure, allowing for calibrations and preventing excessive pressure. Another piece with a cavity is included to secure the marker.

Figure 7 Mechanical Solution Proposal C (Commercial mechanism with 2 DOF)

It is a two-degree-of-freedom mechanism with a metallic structure housing two servomotors for rotational movements on two axes. Ideal for precise orientation.

Analysis of Mechanical Solutions

Solution A: Upon analyzing its feasibility, it was observed that this mechanism exclusively requires the TurtleBot Waffle model, which has a configuration with components distributed over a single level or "floor."
This arrangement has certain limitations due to its spatial geometry, as the robot occupies a larger area in the XY plane, potentially hindering its movement in narrow spaces and increasing the risk of collisions with obstacles.

Solution B: This second mechanism requires the TurtleBot Burger model, which organizes its components across three levels. This configuration offers advantages over Solution A, as the robot occupies less space in the XY plane, facilitating movement through narrow spaces and reducing the risk of collisions.

Solution C: This third mechanism is compatible with both TurtleBot configurations, as it is a commercially available product. It features two degrees of freedom (2 DOF), allowing rotation along the 'z' and 'y' axes. However, to adapt it to hold the marker, some modifications would be necessary.
Its metal structure provides greater durability, although the additional degree of freedom does not offer significant benefits for this application. Moreover, the required adaptations to integrate the marker would involve redesigning the mechanism, which may be less viable and cost-effective.

Table 1 Mechanical Decision Matrix - Conclusion

Solution	Cost	Utility	Durability	Σ	Priority
A	0,0561	0,165	0,0561	0,2772	3
B	0,085	0,25	0,1089	0,4439	1
C	0,0289	0,085	0,165	0,2789	2

The choice of Solution B responds to the need for a system that combines spatial efficiency, ease of integration, and functionality tailored to the purpose. Its optimized design, based on well-distributed components and a controlled pressure system, ensures the best performance for the anticipated conditions. Therefore, it is the most suitable solution for implementation in technical, practical, and economic terms.

Figure 8 Assembled Marker Mechanism

Figure 9 Assembled Marker Mechanism

The implementation of the mechanism will be carried out through 3D printing using silver PETG material (Figure 8 and 9), selected for its strength and finish. The parts to be fabricated include the General Base, Marking Mechanism Base, Rack, Spring Base, Marker Spring Support, Marker, and Pinion. These will be printed with resolutions of 0.2 mm and 0.15 mm, a 15% infill, and supports only in the parts that require them. The total printing time will be approximately 13 hours and 10 minutes, with an estimated cost of $26.50, optimizing quality, precision, and efficiency. To better understand the marker mechanism, it is recommended to watch the assembly video available at [7].

Programming

In the programming section, a modern methodology is implemented that combines advanced tools for image processing, trajectory generation, and precise robot control. This methodology is specifically designed for automated drawing applications using robots, leveraging optimized algorithms to interpret images and translate them into controlled movements.

To test and validate the developed algorithms, advanced tools such as ROS are used, along with its visualization (RViz) and simulation (Gazebo) modules. These tools are essential due to their ability to simulate robot behavior in virtual environments, allowing for dynamic adjustments to system parameters such as speed, precision, and point density in the trajectory.

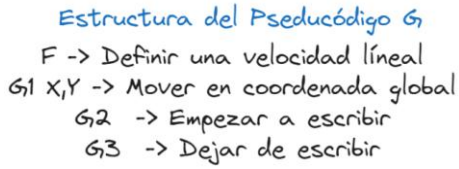

Figure 10 Pseudocode G Proposal

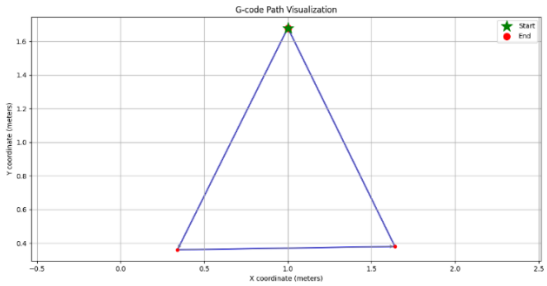

Figure 11 First Version Pseudocode

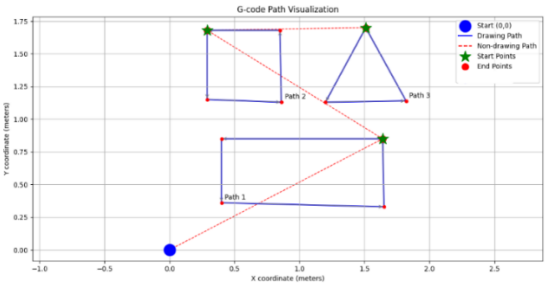

Figure 12 Actual Pseudocode Version

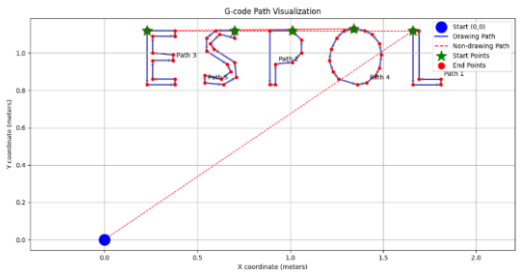

Figure 13 Writting Test

To control the path traced by the robot, a system was developed to generate a pseudocode file with path instructions based on a given image. The pseudocode structure was defined through multiple iterations to meet the project's requirements. The robot's movement is executed by subscribing to the "Odom" topic, using global coordinates as a reference [6].

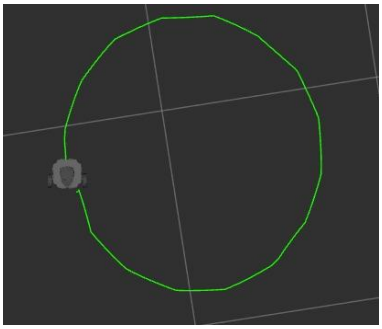

Figure 14 TurtleBot controller simulation

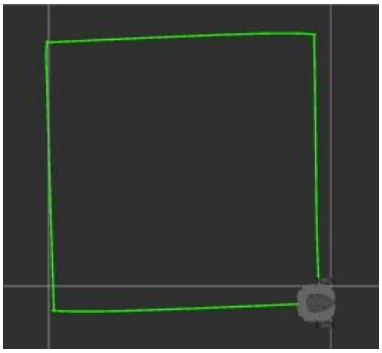

Figure 15 TurtleBot controller simulation

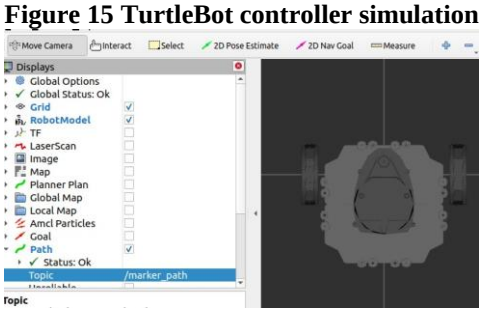

Figure 16 Rviz Configuration

Figures 14 and 15 illustrate the implementation of the code for an advanced TurtleBot controller that interprets and executes trajectories defined in a G-code file. This controller enables the robot to follow specific coordinates while managing a simulated marker to draw lines, ensuring smooth movements through a velocity profile system. Using ROS, the controller manages the robot's pose in real-time, publishes the trajectory to a topic for visualization in RViz, and dynamically adjusts parameters such as speed and orientation to ensure precision and stability.

The functionality includes G-code commands to move the robot (G1) and control the marker (G2 to lower and G3 to raise). It also supports simulations in Gazebo and visualization in RViz, allowing for testing and validating trajectories before physical execution.

It is worth mentioning that the implementation and visualization in simulation is essential because it allows testing and adjusting trajectories safely and efficiently before physical execution, reducing risks and optimizing resources. To better understand the functionality within the simulated environment, it is recommended to watch the video available in reference [8].

Electronic Design

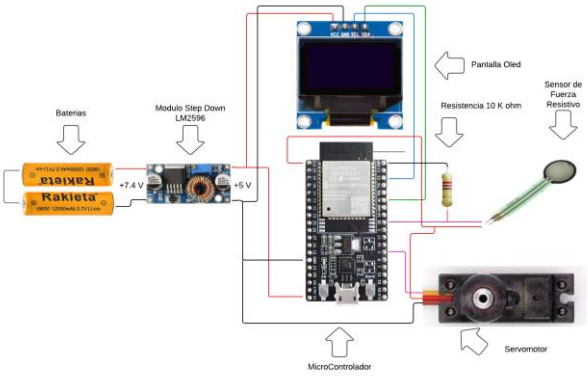

Figure 17 Proposed Electronic Solution

The proposed electronic prototype is an integrated system that utilizes various components to measure force and control a servo motor while displaying data on an OLED screen. The components and their functions are as follows:

- Batteries: The primary power source for the circuit, providing 7.4 V.
- Step-Down Module (LM2596): Reduces the input voltage from the batteries to 5 V, a suitable level for powering the circuit components.
- Microcontroller (ESP32 or Arduino): The "brain" of the system. It coordinates reading the force sensor, controls the servo motor through feedback from the sensor, and manages communication with the OLED screen.
- Force Sensitive Resistor (FSR): Detects the applied force, generating a change in resistance that is read by the microcontroller to calculate the force exerted.
- 10 kΩ Resistor: Part of a voltage divider with the force sensor to adapt the signal for the microcontroller.
- OLED Screen: Displays information such as the force detected by the sensor or the system's status.
- Servo Motor: Controlled by the microcontroller, it performs movements in response to control signals to apply or release force.

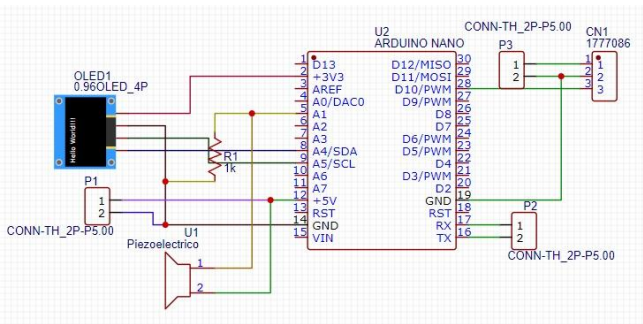

Figure 18 Final Electronic Schematic

The Arduino Nano is an excellent choice for your project due to its compact size, cost-effectiveness, and sufficient features for managing the FSR, servo motor, and OLED display. While the ESP32 offers additional features, they are not required for

this application, and its increased complexity and cost would be unnecessary. The Arduino Nano ensures an optimal balance between functionality, simplicity, and affordability for your force-measuring and servo-controlling prototype.

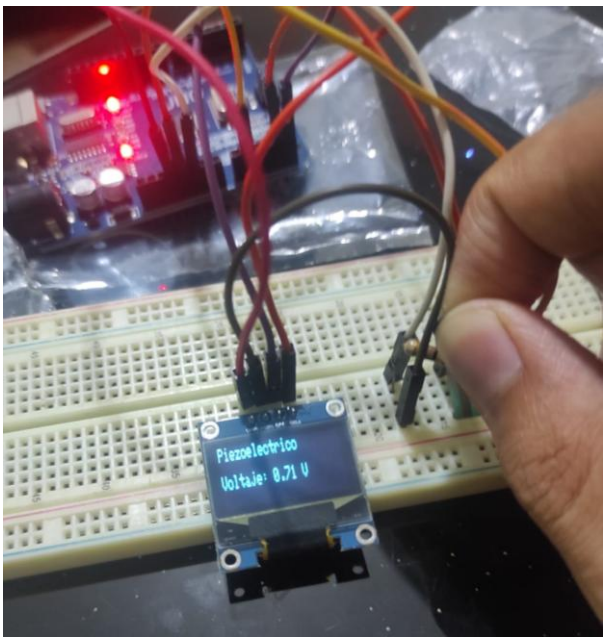

Figure 19 Physically Implementation

Figure 20 Physically Implementation

The circuit was physically implemented to validate the correct functionality of the prototyped schematic shown in Figure 15, achieving satisfactory results. The OLED screen accurately displayed real-time information about the voltage generated by the piezoelectric sensor, as shown in Figure 16, indicating 0.71 V, confirming the correct acquisition and processing of data. Additionally, the servo motor responded appropriately to the voltage provided by the sensor, demonstrating that the control logic and the interaction between components worked as planned. This validates both the functionality and the consistency of the prototyped design and its physical implementation.

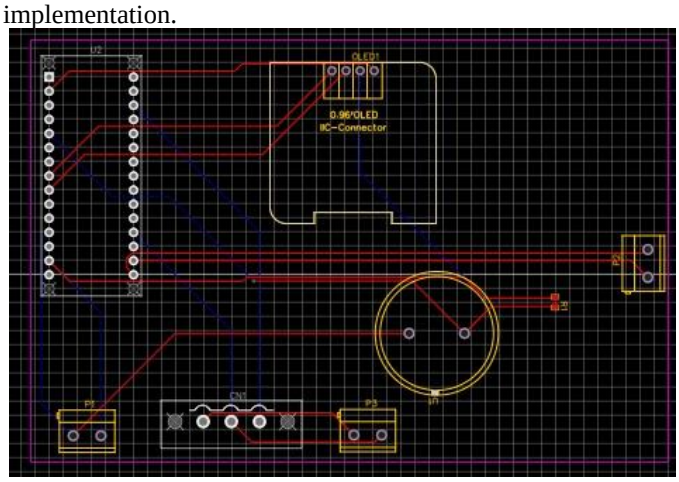

Figure 21 PCB routing for Arduino Nano with connections to peripherals

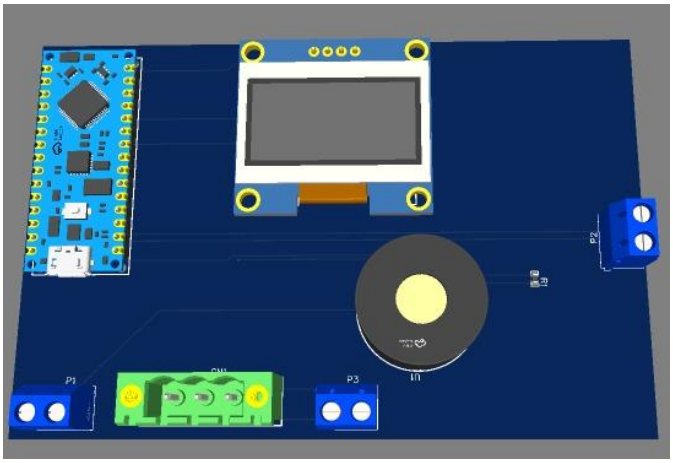

Figure 22 3D View of PCB

The PCB of the circuit was designed after the previous validation process as part of the prototype improvement. This allows for the integration of all components into a compact and organized design with a width of 115 [mm], a height of 76 [mm] and 2 copper layers, optimizing connections and reducing electrical noise present in the previous version with the breadboard. The PCB, shown in Figure 19, facilitates a more stable connection between the Arduino Nano microcontroller, the piezoelectric sensor, the OLED display, and the servo motor, ensuring more reliable and efficient operation.

Mathematical Model

The purpose of this work is to create a writing mechanism that said the cinematic model for the differential robot was not touched and is based on the turtlebot3 package. It uses ROS as the communication framework and the cinematics are based on the following.

$$\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{bmatrix} = \begin{bmatrix}\cos(\theta) & 0\\ \sin(\theta) & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}v\\ \omega\end{bmatrix}$$

Equation 1 Differential cinematic model

With that said we can search for the turtlebot3 manual and write our own code for the odometry, but we can use the one stablished in their repository. We choose to use the second one with ROS framework as mentioned before the the Pseudo G-code.

For our mathematical model, we will use a spring-damper system combined with a servo motor and a pinion-rack mechanism. Using Newton's laws, we will derive the equations for the proposal "B" solution model of the component that will be attached to the TurtleBot3, ensuring it maintains contact throughout its trajectory.

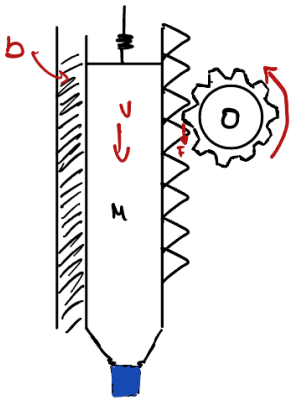

Figure 23 Dynamic Model

From:

$$\boldsymbol{\Sigma\tau = J\ddot{\theta} + \tau_a} \quad \& \quad \boldsymbol{\Sigma F = m\ddot{y}}$$

Equation 2 Newton's Laws

We have:

$$F = m\ddot{y} + b\dot{y} - ky \quad \& \quad \tau_a = F * R$$

Equation 3 Forces and Torques

Where:

F: Force applied
m: Mass
b: Damping between the slider and the marker
k: Spring coefficient
τa: Torque applied
J: Moment of Inertia

We can also say from cinematics that:

$$y = \theta * R$$
$$\dot{y} = \dot{\theta} * R$$
$$\ddot{y} = \ddot{\theta} * R$$

Equation 4 Cinematic equivalents

From these two equations we can establish our model as the following:

$$\ddot{\theta} = \frac{1}{J + mr^2}\left[-br^2\dot{\theta} + kr^2\theta + \tau_{(t)}\right]$$

Equation 5 Dynamic Model

We can represent our model as a state diagram as shown below:

- x1 = θ: Represents the angle of the system, describing the rotational position.
- x2 = θ̇: Represents the angular velocity, which is the rate of change of the angle over time.

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ \frac{-kr^2}{J+mr^2} & \frac{br^2}{J+mr^2}\end{bmatrix}\begin{matrix}x_1\\ x_2\end{matrix} + \begin{bmatrix}0\\ \frac{1}{J+mr^2}\end{bmatrix} * \tau$$

$$y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Equation 2 State Diagram

Even though we did the theoretical model it could also be done if we import the dynamics to MATLAB using simscape, this will give us a better understanding of the plant. We choose this instead of a theoretical approach because we have the designed pieces that will allow us to simulate the model.

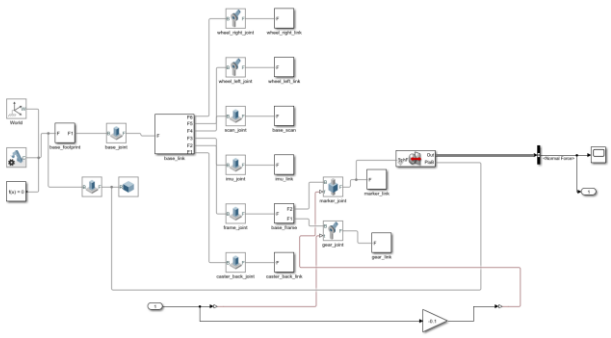

Figure 21 Simscape plant

The system relies on feedback from a piezoelectric force sensor to measure the applied force. This feedback is critical for dynamically adjusting the motor torque to maintain the desired force and correct for disturbances or variations in the surface.

PD controller is the preferred choice to regulate the motor and ensure that the marker maintains consistent contact with the surface while applying the desired force. The decision stems from the need for a system that reacts quickly and precisely to variations in applied force, as the robot operates in dynamic environments where surface irregularities or sudden shifts in force could occur. The proportional term of the PD controller will address the immediate force error by generating a corrective torque proportional to the deviation, while the derivative term will enhance stability by accounting for the rate of change of the error, preventing overshoot or oscillations. The use of the piezoelectric force sensor provides real-time feedback on the applied force, making a PD controller

particularly well-suited to leverage this data for immediate adjustments. [5]

To simulate the sensor, we used the Simscape contact forces library and did the representation on a plane and the marker, this will give us the feedback needed to simulate our controller. We also used the PID block and a PID tuner to impose a behaviour and later test our feedback control.

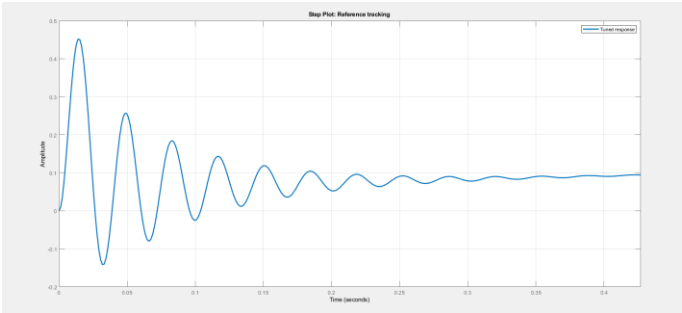

Figure 22 PID tuner

Controller Parameters

	Tuned
P	5.0173
I	10.3478
D	0.1965
N	29556.0945

Performance and Robustness

	Tuned
Rise time	0.00644 seconds
Settling time	5.05 seconds
Overshoot	4.08 %
Peak	1.04
Gain margin	33.5 dB @ 1.48e+03 rad/s
Phase margin	15 deg @ 259 rad/s
Closed-loop stability	Stable

Figure 23 PID gains

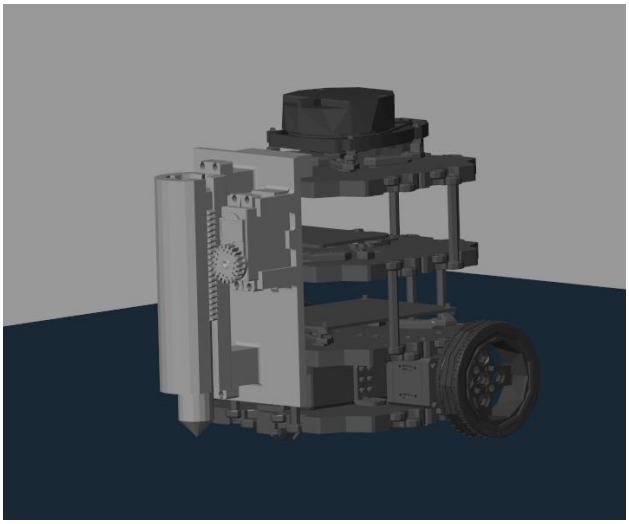

Figure 24 Simulated model

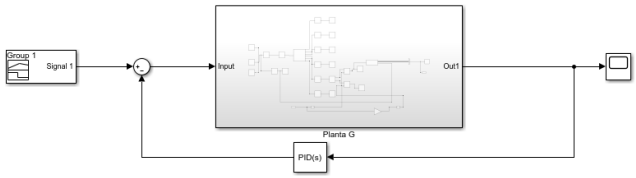

Figure 24 Closed loop model

Lastly we tested our model with the following gains and have some initial results. The setpoint used was a gaussian function created to simulate different irregularities in the plane. This in situations were the marker would have some difficulties and we have the following response from the plant.

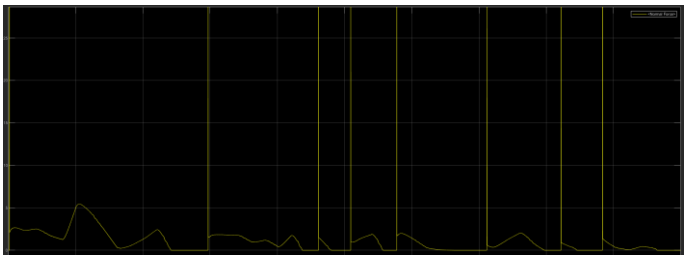

Figure 25 Sensed Force

The peak forces are the worst irregularities that summed up in this impact of the marker on the uneven floor.

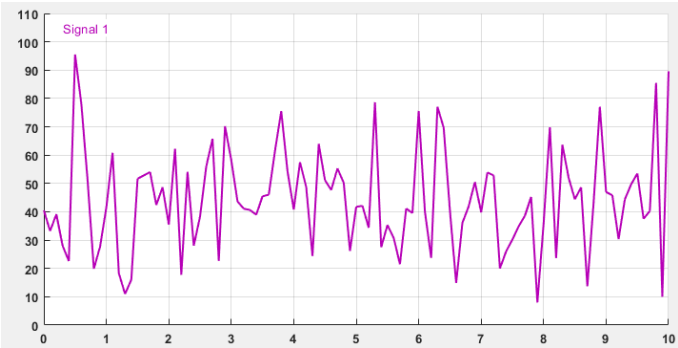

Figure 26 Irregularities

VI. Results and Analysis

The mechanical, electronic, programming, and control design of the system based on the "B" model from Figure 4 was successfully validated through simulations and physical testing. The results indicated that the marking mechanism meets the stated objective, which focuses on the development of an autonomous robot capable of drawing shapes on flat surfaces. This is oriented toward applications in construction and carpentry, aiming to improve precision, reduce errors, and optimize resources through automation. The workflow for using the robot is shown in Figure 27.

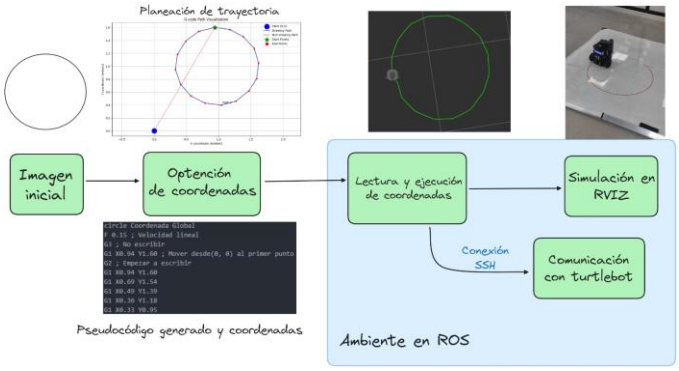

Figure 27 Robot flowchart

The mechanical system is designed to meet the technical specifications required to transform the rotational motion of the servomotor into linear motion using the rack-and-pinion mechanism. The integration of a spring was a key decision in the design, as it allows for the gradual application of force, reducing the likelihood of excessive stresses that could compromise the integrity of the components over time. Additionally, the dimensions and weight of the components were carefully selected to ensure that the kinematics and dynamics of the differential robot TurtleBot3 Burger, shown in Figure 28, remain unaffected. This approach ensures efficient operation and extends the mechanical components' service life.

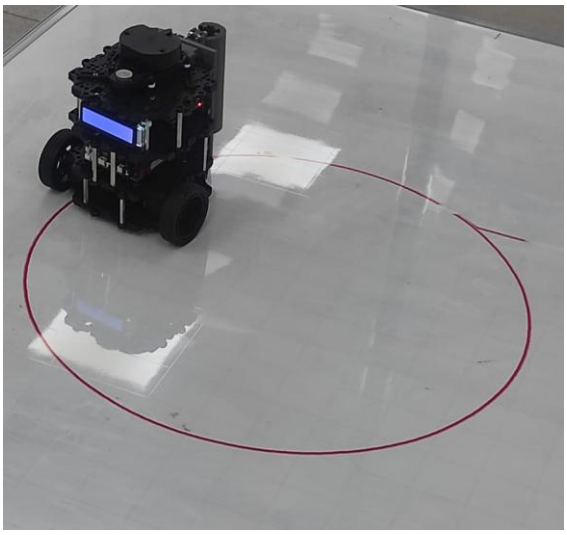

Figure 28 Figure tracing in real application

The design of the electronic and communication system between the Arduino, Raspberry Pi, and peripherals/sensors was developed through an iterative process, which allowed the optimization of programming toward the use of interrupts. These interrupts ensure the immediate detection of signals sent via serial communication between the microcomputer (Raspberry Pi) and the microcontroller (Arduino), achieving low latency in the execution of the figure-drawing process.

Additionally, an I2C communication protocol was integrated to control an OLED display, which is used to visualize in real-time the voltage measured by the piezoelectric sensor. This provides visual feedback to verify the correct behavior of the plant, i.e., the robot equipped with the marking mechanism. To better understand, it is recommended to watch the video available in reference [9].

In terms of hardware design, a printed circuit board (PCB) was avoided because the system connections were simple. Instead, connections were made using direct soldering with copper wires, resulting in a more compact, lightweight, and efficient system in terms of weight and space. This approach was key to maintaining portability and reducing assembly complexity.

The programming results consist of two codes: the first converts PNG images into G-Code trajectories by processing the image, detecting contours, and generating G1, G2, and G3 commands. It also visualizes the generated trajectories to validate their accuracy. The second code uses the G-Code file generated by the first to control a TurtleBot via ROS, interpreting the commands to execute precise and smooth movements while handling a marker for drawing. Both codes work together, enabling complete automation from design to execution, achieving precise drawings, optimized trajectories, and effective visual feedback for both simulations and real-world applications. The real demonstration of figure tracing can be found in reference [10].

For the control of the plant (the robot), a solution based on ROS and Simscape was implemented, integrating a spring-damper system coupled to a servo motor and a rack-and-pinion mechanism. The dynamic model was formulated using Newton's laws, including equations describing forces, torques, and kinematic parameters. This approach, complemented by ROS and Simscape, provides a solid and efficient platform for the development and simulation of complex dynamic systems.

The choice of a PD controller proved to be suitable due to its ability to quickly correct errors and maintain stability in environments with high uncertainty, such as those with uneven surfaces. Initial results confirm that the system responds effectively to force spikes detected by the piezoelectric sensor, demonstrating an optimized design. Furthermore, the proper selection and tuning of the controller gains allowed achieving an ideal balance between fast response and system stability.

This controller was embedded in an Arduino microcontroller, which executes the necessary actions when the Raspberry Pi, acting as the brain of the TurtleBot, sends the signal to lower the marker for drawing the figure. In this process, the controller dynamically adjusts the system to reach a setpoint of 0.55 V, ensuring consistent and precise contact of the marker with the surface during tracing [11].

VII. Conclusions and Recommendations

Conclusions

The selected PD controller proved to be an efficient solution for regulating the marker's contact with irregular surfaces. Its ability to respond quickly to force errors, combined with feedback from the piezoelectric sensor, ensures stability and precision even under dynamic conditions. This approach provides a solid foundation for future implementations in adaptive or robust control.

The use of ROS as a communication and control platform facilitated the integration of advanced simulation tools (Gazebo and RViz) and enabled the development of an automated workflow from design to the physical execution of tracing. This demonstrates ROS's potential for projects requiring modularity and scalability.

The selection of the spring-damper and rack-and-pinion-based mechanism was critical to ensuring the efficient operation of the marker. Its compact design preserved the kinematics of the TurtleBot3, maintaining maneuverability in confined spaces. Additionally, the use of 3D-printed materials balanced cost and mechanical performance, ensuring durability and lightweight construction.

The electronic system design, which included the implementation of an Arduino microcontroller and serial communication with a Raspberry Pi, proved to be functional and reliable. The integration of an I2C protocol for real-time data visualization added an additional layer of system monitoring and validation.

The automated conversion of images to G-Code trajectories and their precise execution using the TurtleBot validated the system's feasibility for practical applications. This automated workflow significantly optimized time and accuracy compared to manual methods, which is critical in sectors such as construction and carpentry.

Simulations in Simscape and physical implementation demonstrated the consistency of the dynamic model and the functionality of the complete system. The use of theoretical models and simulations reduced iteration costs and validated the design before fabrication.

The developed prototype has the potential to transform repetitive and critical tasks in construction and manufacturing. Its precision and efficiency could significantly reduce material waste, rework costs, and production times, improving competitiveness in high-standard projects.

The modularity of the mechanical system and the versatility of ROS make the project scalable to other applications, such as automated cutting systems or more complex industrial tasks, while maintaining a focus on precision and automation.

Recommendations

Including additional sensors such as accelerometers or gyroscopes to gather complementary data on the system's inclination and vibration would enhance stability control and marking precision on more challenging terrains.

Making the marking mechanism interchangeable to accommodate different tools (markers, laser cutters, etc.) would increase the system's versatility for various applications. Additionally, positioning the marking mechanism at the robot's center of rotation would improve symmetry and consistency during operation.

Designing an automatic spring adjustment system to compensate for mechanical wear over time would maintain a constant applied force and eliminate the need for frequent recalibrations.

Integrating navigation algorithms based on SLAM (Simultaneous Localization and Mapping) would enable the robot to operate autonomously in unknown environments, adapt to obstacles, and update trajectories in real time.

Incorporating computer vision algorithms to recognize printed or scanned architectural plans and convert them directly into marking commands would further streamline the workflow and expand the robot's capabilities.

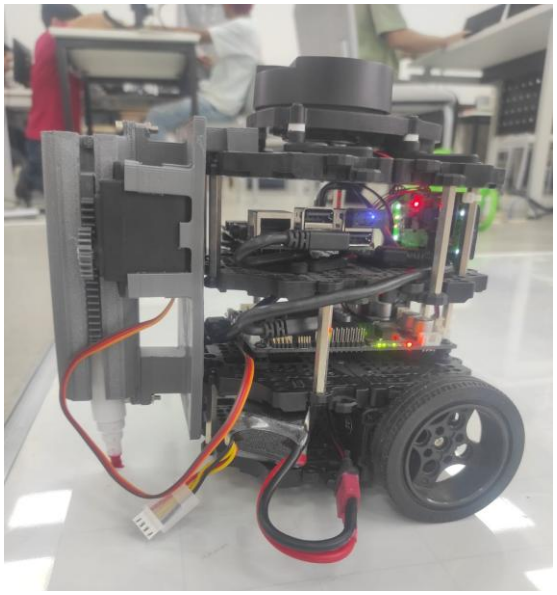

Figure 29 Real Robot

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